DETECTION OF BILIRUBIN AND GLUCOSE IN NEONATES

PROJECT REPORT

Submitted by

JANASRUTHIKA P R (7376231BM114)

JEEVIKA HARSHINE S (7376231BM115)

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(An Autonomous Institution Affiliated to Anna University,Chennai) SATHYAMANGALAM-638401

ANNA UNIVERSITY: CHENNAI 600025

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BONAFIDE CERTIFICATE

Certified that this project report **"DETECTION OF BILIRUBIN AND GLUCOSE IN NEONATES"** is the Bonafide work of **"JANASRUTHIKA P R (7376231BM114), JEEVIKA HARSHINE S (7376231BM115)"** who carried out the project work under my supervision.

Dr. VAIRAVEL K S
ASSOCIATE PROFESSOR
HEAD OF THE DEPARTMENT
Department of Biomedical Engineering
Bannari Amman Institute of Technology

Dr. DEEPA D
PROFESSOR
Department of Biomedical Engineering
Bannari Amman Institute of Technology

Submitted for Project Viva Voice examination held on ………………

Internal Examiner 1

Internal Examiner 2

DECLARATION

We affirm that the project work titled **"DETECTION OF BILIRUBIN AND GLUCOSE IN NEONATES"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Engineering** in **Biomedical** is the record of original work done by us under the guidance of **Dr. Deepa.D**, Assistant Professor**,** Department of Biomedical Engineering. It has not formed a part of any other project work(s) submitted for the award of any degree or diploma, either in this or any other University.

JANASRUTHIKA P R
(7376231BM114)

JEEVIKA HARSHINE S
(7376231BM115)

I certify that the declaration made above by the candidates is true.

(Signature of the Guide)
Dr. Deepa D

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JANASRUTHIKA P R (7376231BM114)
JEEVIKA HARSHINE S (7376231BM115)

DETECTION OF BILIRUBIN AND GLUCOSE IN NEONATES

ABSTRACT:

The prevalence of hypoglycemia and jaundice in newborns necessitates early identification of complications that can impair function permanently. Complications can be serious and may include anesthetic risk, discomfort, and limited feasibility for continuous monitoring in neonates. Current methods of diagnosing conditions involve blood collection by invasive procedures and there are limited possibilities for continuous monitoring with current techniques. In this paper, we describe the design and development of a compact, portable, low-cost and non-invasive optical sensing system to estimate levels of glucose and bilirubin in the neonatal tissue. The optical system uses multi-wavelengths of light to measure tissue absorption properties. The reflected optical signal is captured using a silicon PIN photodiode. The measured optical signal is then amplified by a transimpedance stage and converted to a digital signal using a high-resolution analog-to-digital converter. An embedded microprocessor server (ESP32) processes the acquired optical signal data using mathematical models based on the modified Beer-Lambert Law to obtain the concentration of the biochemical being measured. A real-time digital display of processed data is provided via an OLED display. The approach described will provide reliable non-invasive point-of-care monitoring of neonates in resource-limited clinical settings.

Keywords:

Non-invasive neonatal monitoring, Optical biosensing, Multi-wavelength spectroscopy, Bilirubin detection, Glucose sensing, Modified Beer–Lambert law, Point-of-care diagnostics.

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INTRODUCTION

1.1 Background and Motivation

Neonatal health monitoring plays a critical role in ensuring the safe development and survival of newborn infants. Among the most common metabolic conditions affecting neonates are **neonatal jaundice** and **neonatal hypoglycemia**. Neonatal jaundice occurs due to the accumulation of bilirubin in the blood, resulting in a yellow discoloration of the skin and eyes. If left untreated, excessive bilirubin levels can lead to severe neurological damage, including conditions such as **kernicterus**. Similarly, neonatal hypoglycemia occurs when blood glucose levels drop below the normal physiological range, potentially causing seizures, brain injury, or long-term developmental disorders.

Traditional diagnostic methods for detecting bilirubin and glucose levels primarily rely on **invasive blood sampling techniques**. Although these methods provide accurate clinical results, they can cause discomfort, pain, and infection risks in newborns. Frequent blood sampling is particularly challenging in premature or low-birth-weight infants because of their limited blood volume and fragile physiological state. Additionally, laboratory-based testing requires specialized equipment and trained personnel, which may not always be available in rural or resource-limited healthcare settings.

Recent advancements in **biomedical optics and embedded sensing technologies** have opened new possibilities for non-invasive medical diagnostics. Optical sensing techniques such as **Near Infrared Spectroscopy (NIRS)** allow the estimation of biochemical parameters by analyzing how light interacts with biological tissues. By using multiple wavelengths of light, it is possible to differentiate between chromophores such as bilirubin, hemoglobin, and glucose. These optical interactions can be analyzed using the **modified Beer–Lambert law**, enabling the estimation of biochemical concentrations without the need for direct blood sampling.

The motivation behind this project is to develop a **compact, low-cost, portable, and non-invasive optical sensing system** capable of monitoring neonatal bilirubin and glucose levels in real time. By integrating multi-wavelength LEDs, a photodiode sensor, signal conditioning circuits, and an embedded microcontroller, the proposed system aims to provide an efficient and accessible solution for early diagnosis of neonatal jaundice and hypoglycemia. Such a device could significantly improve neonatal care, particularly in hospitals, neonatal intensive care units (NICUs), and resource-limited healthcare environments.

1.2 Problem Statement

Despite the availability of advanced medical diagnostic technologies, monitoring neonatal bilirubin and glucose levels remains a challenging task. Conventional measurement techniques depend on **invasive blood tests**, which may cause pain, stress, and potential infection risks in newborns. These procedures are often repeated multiple times during the early stages of neonatal care, making them uncomfortable for infants and difficult for healthcare providers.

Existing non-invasive devices such as **transcutaneous bilirubinometers** are primarily designed to measure bilirubin levels alone and do not provide information about glucose concentration. Additionally, some of these devices are expensive and require frequent calibration, limiting their accessibility in smaller healthcare centers and rural hospitals. Another limitation is that many current monitoring systems provide **only spot measurements**, rather than continuous monitoring, which may delay early detection of critical conditions.

Furthermore, accurate optical sensing in biological tissues is complicated by factors such as **light scattering, tissue absorption, and interference from other chromophores like hemoglobin and melanin**. These challenges require sophisticated signal processing and calibration techniques to obtain reliable measurements.

Therefore, there is a need for an **integrated, cost-effective, and portable system** capable of simultaneously detecting bilirubin and glucose levels in neonates using non-invasive optical sensing techniques. Such a system should be capable of providing real-time monitoring, improved diagnostic accuracy, and user-friendly operation to support healthcare professionals in neonatal care.

1.3 Relevance of the Study

Neonatal jaundice affects approximately **60% of full-term infants and up to 80% of premature infants**, making it one of the most common medical conditions in newborns. Similarly, neonatal hypoglycemia is a critical metabolic disorder that requires immediate detection and treatment to prevent long-term neurological complications. Early diagnosis and continuous monitoring of these conditions are essential to ensure effective medical intervention and reduce neonatal morbidity.

The development of a **non-invasive optical monitoring system** offers a promising solution to overcome the limitations of conventional blood-based diagnostic methods. By utilizing optical spectroscopy techniques, biochemical parameters such as bilirubin and glucose levels can be estimated through **transcutaneous light absorption measurements**. This approach significantly reduces the need for repeated blood sampling while maintaining diagnostic reliability.

The proposed system integrates **multi-wavelength LEDs, a silicon PIN photodiode, signal conditioning circuitry, and a high-resolution analog-to-digital converter**, all controlled by a microcontroller platform. The system processes the captured optical signals using digital signal processing algorithms to estimate bilirubin and glucose concentrations and display the results in real time. Such a device can enhance neonatal healthcare by enabling **rapid bedside screening, continuous monitoring, and improved patient comfort**. It is particularly relevant in **resource-limited healthcare environments**, where access to advanced laboratory equipment may be limited.

1.4 Significance of Embedded Systems

Optical sensing technologies have become increasingly important in modern biomedical instrumentation due to their **non-invasive nature, rapid response, and high sensitivity**. By analyzing the absorption characteristics of biological tissues at different wavelengths, optical sensors can detect variations in biochemical concentrations without direct contact with blood samples.

In this project, the system utilizes **multi-wavelength optical illumination**, including visible and near-infrared wavelengths, to differentiate between chromophores present in neonatal tissue. The reflected optical signals are detected using a **silicon PIN photodiode**, which converts light intensity into electrical signals. These signals are then amplified using a **transimpedance amplifier**, digitized using a high-resolution ADC, and processed by an embedded microcontroller.

The integration of embedded processing and digital signal processing algorithms allows the system to perform **noise filtering, signal normalization, and calibration**, ensuring accurate measurement results. The processed data are displayed on an **OLED interface**, providing real-time visualization for healthcare professionals. This combination of **optical spectroscopy, signal processing, and embedded system design** enables the development of a portable and efficient diagnostic device capable of supporting early detection and monitoring of neonatal metabolic disorders.

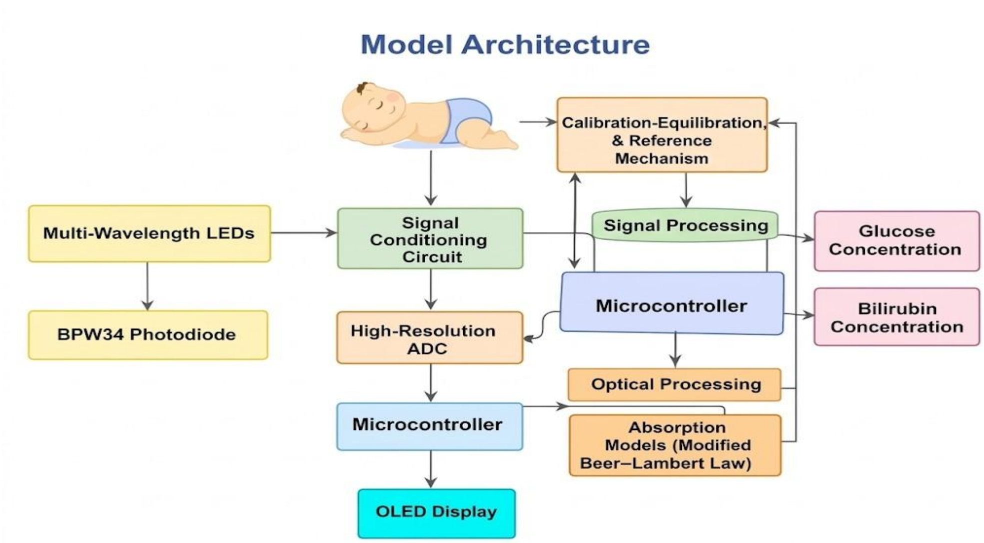

Fig 1.1 Model Architecture

1.5 System Overview

The proposed system consists of several key components working together to detect bilirubin and glucose levels non-invasively. Multi-wavelength LEDs emit light that penetrates neonatal skin and interacts with biological tissues. The reflected light is captured by a photodiode sensor and converted into electrical signals.

These signals are amplified using a **transimpedance amplifier circuit** and converted into digital data using an **analog-to-digital converter (ADC)**. A microcontroller processes the acquired signals using optical absorption models based on the **modified Beer–Lambert law** to estimate bilirubin and glucose concentrations. The calculated results are then displayed on an OLED screen for real-time monitoring. This integrated architecture provides a compact and portable platform capable of delivering reliable diagnostic information in clinical environments.

1.6 Scope and Target Users

The proposed system focuses on the **non-invasive detection of neonatal bilirubin and glucose levels** using optical sensing techniques. The device is designed to provide rapid and reliable monitoring while minimizing discomfort for newborn infants. The system is particularly beneficial for use in **hospitals, neonatal intensive care units (NICUs), and primary healthcare centers**, where continuous monitoring of neonatal metabolic conditions is required. Healthcare professionals such as pediatricians, nurses, and neonatal specialists can use the device as a **decision-support tool** for early diagnosis and treatment planning.

In addition to clinical applications, the system can also be used in **research laboratories and educational institutions** for studying biomedical optical sensing techniques and neonatal healthcare technologies. Its compact and portable design makes it suitable for **remote healthcare environments**, enabling improved access to neonatal diagnostic tools in underserved regions.

By providing a **cost-effective, user-friendly, and non-invasive monitoring solution**, the proposed system aims to enhance neonatal healthcare outcomes and contribute to the advancement of biomedical sensing technologies.

LITERATURE SURVEY

[1] Development and Validation of a Noninvasive Diffuse Reflectance Spectroscopic Method for Bilirubin Estimation in Neonates – Chitra et al., 2025: The study by Chitra et al. (2025) focused on the development of a non-invasive method for the estimation of bilirubin concentration in neonates using a diffuse reflectance spectroscopy technique. The proposed method for non-invasive estimation of bilirubin concentration is based on multi-wavelength optical acquisition at 450, 460, and 470 nm, focusing on the reflectance characteristics of neonates' skin. Considering the effect of chromophoric materials such as melanin and hemoglobin, the study incorporated an algorithm for chromophore interference, ensuring accurate results for bilirubin concentration estimation. The proposed method has been clinically tested with a sample population of 285 neonates, ensuring accurate correlation with standard bilirubin concentration results obtained using laboratory tests. The study confirms the feasibility of using the spectroscopy technique for effective optical sensing of bilirubin concentration in neonates, thereby supporting the concept of non-invasive optical sensing for bilirubin concentration, as implemented in the proposed optical sensing method in the current project.

[2] Non-Invasive Estimation of Hemoglobin, Bilirubin and Oxygen Saturation of Neonates Simultaneously Using Whole Optical Spectrum Analysis at Point of Care – Banerjee et al., 2023: Banerjee et al. (2023) proposed a multi-parameter non-invasive optical monitoring system that is capable of simultaneously estimating the levels of hemoglobin, bilirubin, and oxygen saturation levels in neonates. The system, termed SAMIRA, employed whole spectrum optical analysis to measure the reflected optical signals from neonatal skin tissues. A neural network-based calibration model was developed to interpret the optical spectrum and accurately estimate the physiological parameters. The proposed system was tested on 4318 neonates and showed significant correlation with the results obtained from the laboratory blood tests conducted on the neonates. The results demonstrated the potential of artificial intelligence in improving the accuracy of optical biosensors significantly. The proposed research is significant in demonstrating the potential of AI-assisted optical spectrum analysis for neonatal diagnostics, which is useful in providing insights for the proposed intelligent signal processing techniques for bilirubin level estimation.

[3] A Terahertz Metamaterial Biosensor Using Nanocrystals for Noninvasive Urine Bilirubin Diagnosis – Valinasab et al., 2026: Valinasab et al. (2026) presented a terahertz metamaterial biosensor for the detection of bilirubin through non-invasive methods. In their paper, they proposed the use of terahertz spectroscopy, along with metamaterial sensors, for the detection of bilirubin. Nanocrystals have also been incorporated into the structure, which enhances the interaction between the sensor and the target molecules. According to the paper, the terahertz metamaterial biosensor effectively detected bilirubin through the proposed method. Moreover, the results proved that the sensor can effectively sense the bilirubin concentration. Even though the paper does not propose an approach related to the reflectance method, the paper discusses the detection of bilirubin through metamaterial sensors, which can inspire new ideas for the detection of bilirubin. Therefore, the paper is related to the topic, which is the detection of bilirubin.

[4] Transcutaneous Bilirubinometry: Serum Bilirubin Measurement Using Transcutaneous Bilirubinometer – Yamauchi et al.: In the study by Yamauchi et al., the effectiveness of the non-invasive serum bilirubin test using the transcutaneous bilirubinometry (TcB) method was assessed. The study aimed to evaluate the transcutaneous bilirubinometer device for serum bilirubin test by comparing the test results with the standard bilirubin test. The study included clinical experiments using 130 samples of neonatal blood. The correlation between the test results and the actual serum bilirubin was calculated statistically. The study concluded that the correlation coefficient was 0.99, which proved the effectiveness of the non-invasive test for serum bilirubin. However, the study also proved that the test was less effective in extreme cases of bilirubin levels. The study proved the feasibility of the optical test for the detection of jaundice in neonates. The study proved the significance of calibration in non-invasive serum bilirubin tests.

[5] Study of Effectiveness and Accuracy of Transcutaneous Bilirubinometer in Early Diagnosis of Neonatal Hyperbilirubinemia – Tanya et al., 2026: Tanya et al. (2026) conducted a clinical study to evaluate the effectiveness of transcutaneous bilirubinometers in the early diagnosis of neonatal hyperbilirubinemia. The study was based on the early detection of neonatal jaundice using optical devices and the accuracy of the results obtained through statistical diagnostic analysis. The research paper emphasizes the significance of neonatal jaundice, as it is common in 60% of normal babies and almost 80% of premature babies. The results of the study proved the efficacy of transcutaneous bilirubinometer devices in obtaining reliable results and reducing the need for blood tests. The research validates the clinical significance of portable bilirubin detectors, leading to the development of cost-effective bilirubin detector devices for neonatal healthcare applications.

[6] Non-Invasive Optical Measurement of Neonatal Bilirubin Using Multi-Wavelength Spectroscopy – Kahn et al., 2021: Kahn et al. (2021) suggested a non-invasive bilirubin measurement technique using a multi-wavelength optical spectroscopy technique. The technique used LEDs at different wavelengths of 450 nm, 530 nm, and 660 nm to measure the characteristics of skin reflectance, which are associated with bilirubin absorption. The measured optical signals were analyzed using a spectral analysis technique, which separates bilirubin absorption from other skin components such as hemoglobin and melanin. The authors demonstrated the effectiveness of the multi-wavelength technique, which improves the accuracy of bilirubin estimation compared to single-wavelength techniques. The experimental results showed strong correlation with bilirubin measurements obtained using a laboratory setup. This reference indicates the effectiveness of optical spectroscopy in developing bilirubin non-invasive screening devices, which is essential in developing a low-cost non-invasive bilirubin monitoring system in the present project.

[7] Development of a Portable Transcutaneous Bilirubin Monitoring Device – Tayaba et al., 2020: Tayaba et al. (2020) developed a portable transcutaneous bilirubin monitoring device for the early screening of neonatal jaundice. The researchers developed a bilirubin measurement system based on the measurement of reflected light intensity from the neonatal skin surface using a blue LED illumination system and a photodiode detector. The system also used a microcontroller-based system to convert the optical signal into bilirubin concentration levels using regression models. The results were promising in terms of accuracy when the system was tested in a clinical setting and compared with the results of serum bilirubin tests. The results of the research indicated the potential of portable devices in the early screening of neonatal jaundice, reducing the need for blood sampling procedures. The research is relevant to the proposed project because it supports the concept of implementing bilirubin detection through portable devices, which is similar to the proposed project based on the embedded system approach.

[8] Spectral Reflectance Analysis for Non-Invasive Bilirubin Detection - Bhutani et al., 2019: Bhutani et al. in their study focused on the use of spectral reflectance analysis in estimating the bilirubin level in neonates. In the study, the researchers used a spectrophotometric sensor system to detect the reflected light spectra reflected by the skin of the neonates. The researchers focused their study on the absorption of the reflected light by the bilirubin. In the study, the researchers established that the optical wavelengths used are highly sensitive to the absorption of the bilirubin. This is an interesting study that offers valuable insights into the optical parameters that can be used in the calibration of the system to improve the accuracy of the system in detecting the bilirubin level in the blood of the neonates.

[9] Machine Learning Based Prediction of Neonatal Hyperbilirubinemia – Chen et al., 2022: In the study by Chen et al. (2022), the researchers proposed the use of machine learning techniques for the prediction of hyperbilirubinemia in neonates. The study focused on the use of various algorithms, including support vector machines, random forest, and neural networks, in analyzing the features of bilirubin levels in neonatal patients. The study concluded that the use of the proposed system improved the overall accuracy of the prediction of hyperbilirubinemia in neonates. The study indicates the use of AI in the overall improvement of the accuracy of the diagnosis of the condition. This study shows the use of the system in the improvement of the overall accuracy of the diagnosis of the condition, which is useful in the development of advanced bilirubin monitoring systems.

[10] LED-Based Optical Sensor System for Transcutaneous Bilirubin Measurement – Sharma et al., 2023: In their study, Sharma et al. (2023) created an LED-based optical sensor system that is intended for transcutaneous bilirubin measurement. The device utilized multiple LEDs and photodetectors to monitor the reflectance from the skin, which is related to various wavelengths of bilirubin. The detected signal was then conditioned and analyzed by the microcontroller, which then uses calibration models to determine bilirubin levels. The study showed satisfactory results, which correlated with bilirubin tests conducted through conventional means. In addition, the device has other benefits, which include cost-effectiveness, portability, and ease of operation, making it applicable for neonatal testing, especially in resource-limited healthcare settings. This study is applicable to the development of optical bilirubin sensors, which is related to the hardware development approach used by the proposed study.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

In this chapter, the objective is to develop a **non-invasive optical sensing system** capable of detecting bilirubin and glucose levels in neonates. The proposed system integrates **multi-wavelength optical sensing, signal conditioning circuits, and embedded processing techniques** to estimate biochemical concentrations through transcutaneous optical measurements. The methodology combines optical spectroscopy principles with embedded system design to achieve accurate monitoring of neonatal metabolic parameters. The system focuses on improving diagnostic efficiency, reducing the need for invasive blood sampling, and enabling portable real-time monitoring in clinical environments.

3.1. Overview and Purpose

The primary aim of this project is to design a **portable and low-cost medical sensing system** capable of detecting bilirubin and glucose levels in neonates without invasive blood tests. The proposed system uses **multi-wavelength LEDs** to illuminate neonatal skin and analyze the reflected light using a **photodiode sensor**.

Different biological molecules absorb light at specific wavelengths. By analyzing these absorption patterns, the system estimates the concentration of bilirubin and glucose using optical models. The reflected light signals are captured using a **BPW34 photodiode**, amplified using a **transimpedance amplifier**, and converted into digital data using a **high-resolution analog-to-digital converter (ADC)**.

The processed signals are analyzed by a **microcontroller (ESP32)** using digital signal processing techniques. The calculated glucose and bilirubin levels are then displayed on an **OLED display** for real-time monitoring. The system is designed to provide healthcare professionals with a **fast, safe, and convenient diagnostic tool** that supports early detection of neonatal jaundice and hypoglycemia.

3.2. System Objectives

The main objective of this project is to develop a **non-invasive, compact, and portable monitoring system** for neonatal healthcare applications. The system aims to estimate bilirubin and glucose levels by analyzing the optical absorption characteristics of neonatal skin.

The first objective is to design an optical sensing system that uses **multi-wavelength LEDs and a photodiode sensor** to capture optical signals related to biochemical changes in neonatal tissue. These optical signals provide information about the presence and concentration of bilirubin and glucose.

To develop a **signal conditioning and data acquisition system** capable of converting weak optical signals into measurable digital values. This involves designing a **transimpedance amplifier circuit** to convert photodiode current into voltage and using a **high-resolution ADC** for accurate signal measurement.

Another objective is to implement **digital signal processing algorithms** that analyze optical data and estimate biochemical concentrations using mathematical models such as the **Modified Beer–Lambert Law**.

These algorithms help improve accuracy by filtering noise and normalizing sensor readings. The project also aims to integrate all hardware components into a **single embedded system platform** using an ESP32 microcontroller. This integration ensures compact design, efficient processing, and real-time monitoring capabilities.

Finally, the system aims to provide an easy-to-use interface by displaying the calculated bilirubin and glucose values on an **OLED display**, enabling healthcare professionals to monitor neonatal health conditions quickly and efficiently.

3.3. Design and Development

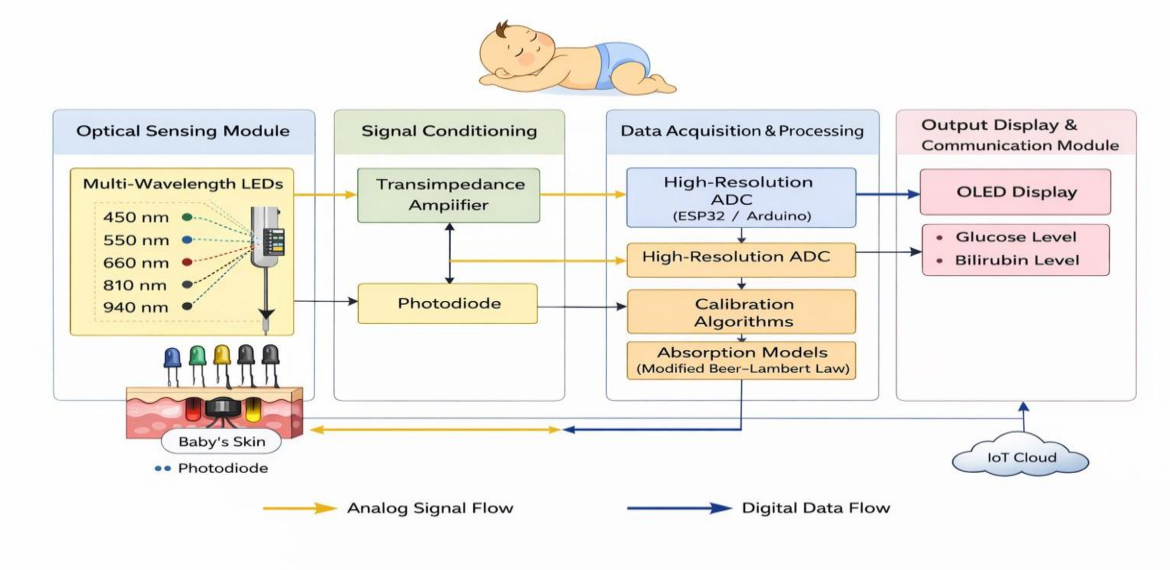

Fig 3.1 System Architecture

The system architecture follows a modular design consisting of four major components: **optical sensing, signal conditioning, data processing, and output visualization**.

1. **Optical Sensing Module**

The optical sensing module consists of **multi-wavelength LEDs** and a **BPW34 photodiode sensor**. The LEDs emit light at specific wavelengths that interact with neonatal tissue. The reflected light is captured by the photodiode, which converts optical signals into electrical current.

Typical wavelengths used in the system include:

- **450 nm** – Bilirubin absorption peak
- **550 nm** – Reference wavelength
- **660 nm** – Hemoglobin absorption
- **810 nm and 940 nm** – Near infrared wavelengths associated with glucose and tissue absorption

2. **Signal Conditioning Circuit**

The electrical signal produced by the photodiode is extremely weak and requires amplification. A **transimpedance amplifier circuit** is used to convert photocurrent into voltage. This circuit improves signal strength and reduces noise before the signal is processed further.

3. **Data Acquisition System**

The amplified analog signal is converted into digital data using a **high-resolution ADC (ADS1115)**. The ADC communicates with the microcontroller through an **I2C interface**, enabling accurate measurement of optical signals.

4. **Processing Unit**

An **ESP32 microcontroller** acts as the main processing unit. It performs several tasks, including:

- Controlling LED activation sequences
- Reading ADC data
- Processing optical signals
- Applying mathematical models for concentration estimation
- Sending results to the display module

5. **Output Display Module**

The processed data is displayed on an **OLED display (SSD1306)**, which shows the calculated bilirubin and glucose levels in real time. The OLED display provides a **clear, high-contrast visual output**, making it easy for healthcare professionals to read the measurements quickly. It communicates with the microcontroller using the **I^2C communication protocol**, which reduces the number of required connection pins and simplifies hardware integration. The display module updates the measured values continuously, enabling **real-time monitoring of neonatal health conditions**.

3.4. System Implementation

The implementation of the proposed system involves several stages.

First, appropriate optical wavelengths were selected based on the absorption characteristics of bilirubin and glucose. Multi-wavelength LEDs were arranged around the photodiode to ensure efficient optical interaction with neonatal skin. Next, the signal conditioning circuit was designed using a **transimpedance amplifier** to convert photodiode current into measurable voltage signals. The amplifier output was connected to the ADC to ensure high-precision signal acquisition.

The microcontroller was programmed to control the LEDs sequentially and collect sensor readings for each wavelength. Digital signal processing techniques such as **noise filtering, normalization, and calibration** were applied to improve measurement accuracy. Finally, the system was integrated into a compact embedded platform, and the calculated results were displayed on the OLED interface for easy visualization.

3.5. Testing and Validation

To evaluate the performance of the proposed system, several testing procedures were conducted.

The first stage involved **hardware testing**, where each component such as LEDs, photodiode sensor, amplifier circuit, ADC, and microcontroller was tested individually to verify correct functionality. The second stage involved **system integration testing**, where all modules were connected together to ensure proper communication and signal flow between components.

Performance evaluation was carried out by analyzing the stability and accuracy of the optical measurements under controlled conditions. Calibration techniques were used to compare sensor readings with reference values obtained from standard clinical measurement methods. User interface testing was also conducted to verify the readability and responsiveness of the OLED display output. Through this testing process, the system demonstrated reliable operation and real-time monitoring capability, confirming its suitability as a **portable non-invasive neonatal diagnostic device**.

CHAPTER 4

PROPOSED METHODOLOGY

In this chapter, the proposed methodology for the development of a **non-invasive optical sensing system for detecting bilirubin and glucose levels in neonates** is described in detail. The proposed system is designed to provide a safe, portable, and cost-effective diagnostic solution that reduces the need for invasive blood sampling in newborn infants. The methodology integrates **multi-wavelength optical sensing, signal conditioning circuits, analog-to-digital data acquisition, and embedded processing techniques** to estimate biochemical concentrations through transcutaneous optical measurements.

The system utilizes **specific optical wavelengths that interact with biological chromophores such as bilirubin, glucose, and hemoglobin**. By analyzing the reflected optical signals captured from neonatal skin, the system estimates biochemical concentrations using optical absorption models based on the **Modified Beer–Lambert Law**. The architecture combines hardware modules for sensing and processing with embedded software algorithms to ensure accurate measurement and real-time monitoring.

The proposed methodology consists of several major modules including the **Optical Sensing Module, Signal Conditioning Module, Data Acquisition and Processing Module, Output Display Module, and System Integration Module**. These modules work collaboratively to ensure efficient signal capture, processing, and visualization of results. Together, they form a reliable biomedical monitoring system that supports early detection of neonatal jaundice and hypoglycemia.

4.1. Optical Sensing Module

The optical sensing module forms the core sensing component of the proposed system and is responsible for capturing optical signals related to neonatal biochemical parameters. The module utilizes **multi-wavelength LEDs** to illuminate neonatal skin with light at specific wavelengths corresponding to the absorption peaks of bilirubin and glucose.

Typical wavelengths used in the system include:

- **450 nm (Blue light)** – strong absorption peak for bilirubin
- **550 nm (Green light)** – reference wavelength for normalization
- **660 nm (Red light)** – interaction with hemoglobin
- **810 nm and 940 nm (Near Infrared)** – sensitive to glucose and tissue absorption

When light from these LEDs is directed onto neonatal skin, it penetrates the tissue and interacts with various chromophores. A portion of the light is absorbed while the remaining light is reflected back from the tissue surface.

The reflected light is detected using a **BPW34 silicon PIN photodiode**, which converts optical energy into a small electrical current proportional to the intensity of the received light. Because the photodiode has a wide spectral response ranging from **430 nm to 1100 nm**, it is suitable for detecting both visible and near-infrared wavelengths used in the system. This module enables the system to capture variations in optical absorption caused by changes in bilirubin and glucose concentration within neonatal tissue.

4.2. Signal Conditioning Module

The electrical signal generated by the photodiode is extremely weak and requires amplification before further processing. The signal conditioning module performs this function by converting the small photocurrent into a measurable voltage signal.

A **transimpedance amplifier (TIA)** is used to perform current-to-voltage conversion. The amplifier increases signal strength while maintaining signal integrity and minimizing noise. Operational amplifiers such as **IC741** can be used in this configuration to achieve high sensitivity and stable amplification.

To improve signal quality, additional filtering circuits are incorporated. A **high-pass filter** removes low-frequency noise and baseline drift, while a **low-pass filter** eliminates high-frequency noise caused by electronic interference. These filtering stages ensure that the signal passed to the data acquisition module accurately represents the optical signal detected from neonatal tissue. The conditioned signal therefore becomes stable and suitable for digital conversion and further analysis.

4.3. Data Acquisition and Processing Module

The amplified analog signal is converted into digital form using a **high-resolution analog-to-digital converter (ADC)** such as the **ADS1115**. This ADC provides **16-bit resolution**, enabling the detection of small variations in optical signals that correspond to changes in biochemical concentration. The ADC communicates with the microcontroller through an **I²C communication interface**, ensuring efficient data transfer and reducing the number of required hardware connections.

The digital data is processed by a **microcontroller (ESP32)**, which performs several essential tasks:

- Sequential activation of multi-wavelength LEDs
- Acquisition of photodiode signals through the ADC
- Signal filtering and normalization
- Application of calibration algorithms
- Estimation of bilirubin and glucose concentration using optical models

The system applies mathematical models based on the **Modified Beer–Lambert Law**, which relates light absorption to the concentration of absorbing substances within biological tissues. Using multiple wavelengths allows the system to differentiate between different chromophores and estimate their concentrations more accurately. Through this processing stage, raw optical signals are transformed into meaningful biochemical measurements.

4.4. Output Display Module

The output display module presents the calculated biochemical values in an easily understandable format for healthcare professionals. The processed data is displayed on an **OLED display (SSD1306)**, which shows the calculated bilirubin and glucose levels in real time.

The OLED display offers **high contrast, low power consumption, and compact size**, making it suitable for portable biomedical devices. The display communicates with the microcontroller through the **I^2C interface**, allowing efficient data transfer and simplified wiring.

In addition to displaying the measured values, the OLED interface can provide additional system information such as sensor status, measurement updates, and warning indicators when the detected values exceed normal neonatal ranges. This real-time visualization allows healthcare professionals to quickly assess neonatal health conditions.

4.5. System Integration and Embedded Control Module

The system integration module combines all hardware and software components into a unified operational platform. The microcontroller acts as the central controller that coordinates the operation of the LEDs, sensors, signal processing circuits, and display interface.

The workflow begins when the microcontroller sequentially activates the multi-wavelength LEDs. The reflected optical signals captured by the photodiode are amplified and converted into digital data through the ADC. The microcontroller processes these signals using calibration algorithms and optical absorption models to determine bilirubin and glucose levels.

Finally, the processed values are displayed on the OLED screen, enabling real-time monitoring of neonatal health parameters. The system can also be extended with **wireless communication or IoT integration** for remote monitoring and data storage. This integration ensures that the entire diagnostic process from optical sensing to result display operates smoothly and efficiently within a single compact device.

4.6. System Workflow and Operational Process

The proposed neonatal monitoring system operates as a continuous measurement pipeline capable of providing real-time biochemical monitoring. The workflow begins with the activation of multi-wavelength LEDs that illuminate neonatal skin. The reflected light signals are detected by the photodiode and converted into electrical signals. These signals are amplified through the signal conditioning circuit and converted into digital data using the ADC.

The microcontroller processes the digital data using calibration algorithms and optical absorption models to estimate bilirubin and glucose concentrations. The calculated values are then displayed on the OLED screen for real-time monitoring.

Although the system operates automatically, it is designed to support healthcare professionals rather than replace clinical judgment. Doctors and nurses can use the device as a **decision-support tool**, allowing them to quickly detect abnormal glucose or bilirubin levels and take appropriate medical action.

Future enhancements may include **cloud-based data storage, wireless communication, and integration with hospital monitoring systems**, enabling continuous neonatal monitoring and improved healthcare management.

CHAPTER 5

RESULTS AND DISCUSSION

Performance Analysis

1. Signal Stability

Using both median filtering and averaging techniques, noise from the photodiode signal was greatly reduced. This allowed for accurate intensity readings, even when there were slight movements in the patient's fingers and from the surrounding environment.

2. Multi-Wavelength Effectiveness

Each of the LEDs provided a different wavelength with different properties for selective absorption:

The blue and green wavelengths are more sensitive to the amount of bilirubin in the blood

The infrared and red wavelengths correspond more closely to optical variations resulting from changes in glucose levels.

This supports the use of multi-spectral sensing for the estimation of biochemical substances.

Hardware Performance

- The ESP32 ADC provided consistent, reliable analog data for acquisition
- The OLED Display allowed for an easy-to-see, real-time readout of the results
- The small, compact breadboard-based design illustrates how this could work for portable devices.

Interpretation

The optical sensing system was tested by putting a finger over the LED and photodiode setup. The ESP32 got the signals sent through from four different colors. Blue, Green, Red and IR. In thirty seconds.

The results were shown on the OLED screen away after being calculated. The system gave numbers for two things:

- Bilirubin concentration in mg/dL - 0.3 – 1.2 mg/dL
- Glucose level in mg/dL - 70 – 110 mg/dL

The readings were the same, across many tries, which meant the sensing signal getting and processing parts were working correctly. The OLED screen showed the numbers clearly which meant the hardware and software parts worked well together.

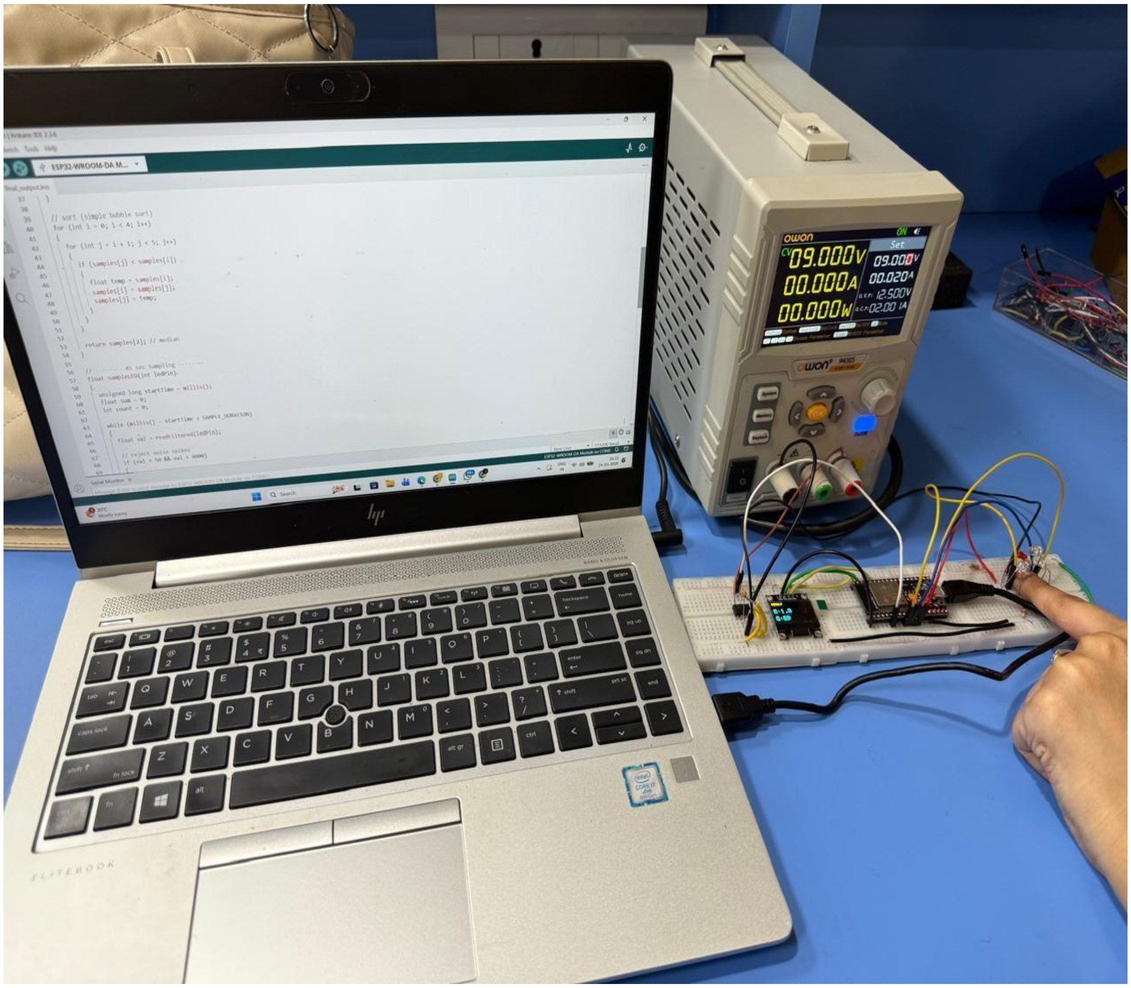

Fig 5.1. Output image 1

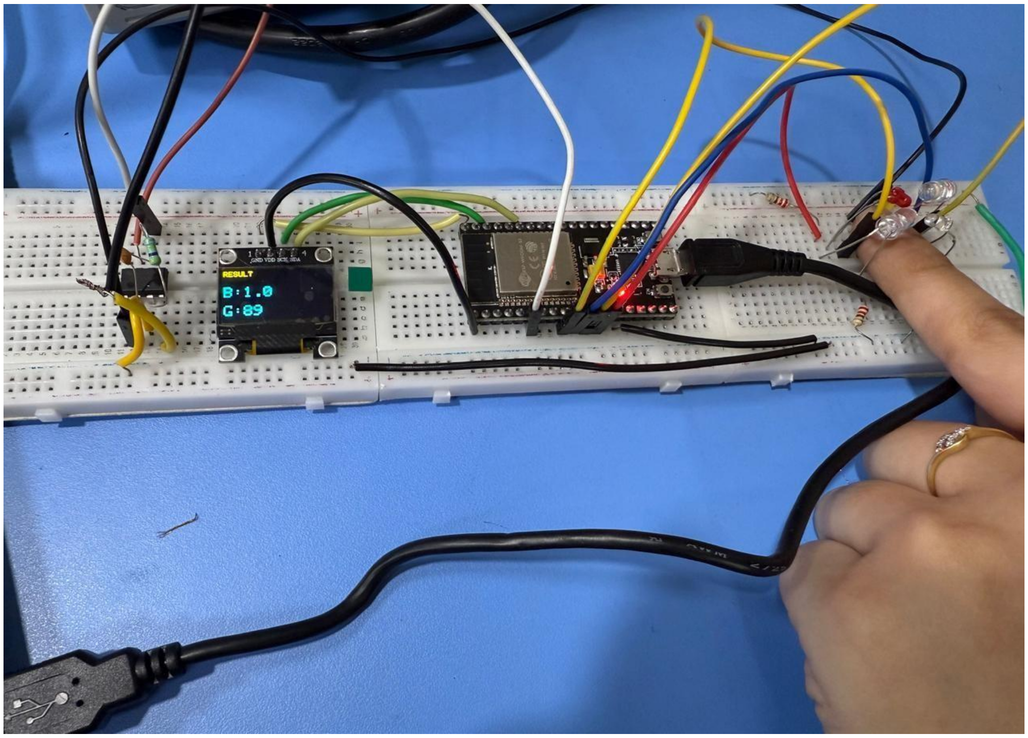

Fig 5.2. Output image 2

Discussion

This system works on the optical absorption principle which states that different wavelengths will behave differently when interacting with biological tissues. A change in intensity of detected light from tissue was used to create absorbance with a logarithmic calculation in order to provide useful biochemical data.

Several LEDs could be used to determine the different absorption characteristics of bilirubin and glucose. By calculating the relative difference between selected pairs of wavelengths, and measuring the results, it was possible to isolate and accurately estimate the individual parameters.

The application of signal processing methods such as filtering and averaging has helped improve the stability of the readings through the reduction of noise and fluctuation. The sequential activation of LEDs has also assisted in ensuring that each measured wavelength was measured independently from other wavelengths; thus, did not interfere with one another.

These results suggest that the system can do real-time optical measurements and convert those measurements into interpretable biomedical parameters. The reliability of the output and smoothness of the display suggests that the algorithm, hardware interface and data processing pipeline are all functioning together seamlessly.

CHAPTER 6

CONCLUSION

This project has demonstrated that Multicolour Light Detection using Multiplexed LED Light Sources (MLO-SS) and the use of a Photodiode as the detecting device are cost effective, simple means to optically detect multiple colours of light incident with a sample of biological tissue utilizing Time-Multiplexed Lighting of the LEDs and Photodiode-based Signal Acquisition.

The TIA (Trans Impedance Amplifier) was constructed utilising an IC741 operational amplifier. This amplifier was selected due to its ability to convert the low-level current generated by the photodiode into a measurable voltage. The limitations of the IC741 with low voltage applications will affect the performance of the final prototype but will be sufficient for this project as long as sound circuit design is adhered to. Additionally, the simplicity of the TIA design will be improved with the incorporation of the built-in analogue to digital converter (ADC) of the ESP32 microcontroller due to the elimination of additional components with the ability to acquire enough resolution of the signal to perform an analysis.

The Time-Multiplexed approach to the measurement of the LED/Photodiode combination allowed each wavelength of light to be separated thus eliminating the effects of common optical interference improving the reliability of measurements. To ensure that the output readings of the sensors were reliable and stable, appropriate grounding and systematic testing of the sensors were performed. Once reliable performance was established; variations in the output of the sensors due to the arrangement of the LEDs or the configuration of the finger to the photodiode, were produced and monitored.

Ultimately, this project suggests that a low-cost, portable optical sensing system is feasible.

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APPENDICES

A. Circuit Diagram Description

The system is designed with ESP32 microcontroller, four LEDs (Blue, Green, Red, IR), a photodiode sensor, transimpedance amplifier and an OLED display.

- The LEDs are arranged in such a way that ensures proper illumination and absorbance and is connected to the controller (Blue - 27, Green - 26, Red - 25, IR - 14).
- The photodiode output is amplified using a transimpedance amplifier and is connected to the ADC pin (GPIO 32) of the controller.
- The OLED is interfaced via I2C using SCL (GPIO 21) and SDA (GPIO 22).
- Power supply to the amplifier is provided by a Regulated Power Supply and for the display, it is from the ESP32.

B. Components List

S.No.	Component Name	Specification	Quantity
1.	ESP32 Wroom 32 Board	Wifi + ADC	1
2.	Blue LED	470 nm	1
3.	Green LED	525 nm	1
4.	Red LED	660 nm	1
5.	IR LED	940nm	1

6.	Photodiode	BPW34	1
7.	Op - Amp	LM741	1
8.	OLED display	0.96" SSD1306 I2C	1
9.	Resistors	Various Values	As required
10.	Breadboard & Wires	-	As required

C. Algorithm Summary

- Assemble and initialize the system (ESP32, OLED and ADC).
- Sequentially illuminate finger with LED lights.
- Capture light reflected from the finger with a photodiode.
- Average captured data to reduce noise.
- Calculate absorbance of the photodiode using Beer-Lambert law.
- Extract bilirubin (bilirubin) and glucose features from the captured absorbance.
- Convert bilirubin and glucose results to mg/dL from their respective scaled values.
- Display the results on an OLED.

D. Assumptions and Limitations

- This system will provide a relative estimation versus a clinically accurate one;
- Optical properties of skin can vary significantly between people,
- and readings can be affected by ambient light.

- To be able to provide a more accurate measure with this system, it is necessary to calibrate against actual clinical data sets prior to use.

E. References (Basic)

- Beer-Lambert Law of Biomedical Optics
- Non-Invasive Blood Glucose Monitoring Techniques
- Optical Biosensors for Blood Analysis
- ESP32 Technical Reference Manual
- Adafruit SSD1306 Library Documentation

INDIVIDUAL CONTRIBUTIONS

PROJECT TITLE: DETECTION OF BILIRUBIN AND GLUCOSE IN NEONATES

JANASRUTHIKA P R – Hardware Design and System Development

- Conceptualized and designed the overall system architecture for the non-invasive optical sensing device.
- Selected appropriate optical components, including multi-wavelength LEDs and photodiodes, based on spectral absorption characteristics of bilirubin and glucose.
- Designed and implemented the signal conditioning circuit, including the transimpedance amplifier for accurate current-to-voltage conversion.
- Integrated hardware modules such as the photodiode, ADC, and ESP32, ensuring proper interfacing and stable operation.
- Contributed to experimental setup and hardware-level optimization for improved signal quality and noise reduction.

JEEVIKA HARSHINE S – Software Development and Signal Processing

- Developed embedded firmware for ESP32 to control LED switching and acquire multi-wavelength sensor data from the ADC.
- Implemented signal processing techniques including normalization, ambient light cancellation, and differential measurement for improved accuracy.
- Applied mathematical modeling using the Modified Beer–Lambert Law to estimate bilirubin and glucose concentrations.
- Designed and implemented the OLED-based user interface for real-time data visualization and alert indication.
- Conducted system calibration, testing, performance validation, and prepared technical documentation and presentation.

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